

Favorability of Behavior Change Techniques and Co-walking Habits Among

Sofia Roth, Luis Jordan, BS, Ravneet Sahni, MS, Danielle Miller, MPH, Ashley M. Goodwin, PhD, Institute of Health System Science

BACKGROUND

- AD/ADRD are increasing with an aging population, impacting caregivers by reducing their quality of life and their emotional and physical health.
- AD/ADRD caregivers often experience high levels of sedentary behavior and low physical activity, both of which are independently associated with an increased risk of mortality. Few behavioral interventions have sought to increase physical activity in AD/ADRD caregivers.
- Behavior Change Techniques (BCTs) are the active ingredients of behavior change interventions and are empirically supported for promoting health behaviors, including physical activity, and may be an effective strategy to implement in text-based walking interventions.

OBJECTIVES

Overall: Examine AD/ADRD caregiver experience with intervention components and impact on care recipients.

RQ1a: Which BCT is most favorably rated by AD/ADRD caregivers?

- **H1a:** Prompt/Cues BCT will be rated most favorably.
- RQ1b:** Did BCT favorability ratings differ between the daily walking habit intervention (1-hour Trial) and the hourly walking habit intervention (4-hour Trial)?
- **H1b:** BCT favorability ratings will be significantly higher in the 1-hour Trial due to lower message burden.
- RQ2:** Is the frequency of co-walking associated with caregiver-perceived improvements in care recipients' health?
- **H2:** Predicted higher co-walking frequency will be positively associated with the care recipient's physical and emotional health.

METHODS

Study Design

AD/ADRD caregivers were enrolled and assigned to one of two text-based walking interventions: a 1-hour walking trial to increase daily step counts by 1,000 steps (N=69) and a 4-hour walking trial to increase daily step counts by 250 steps/hour for a 4-hour timeframe (N=77). Four BCTs were delivered daily via text message for 10 weeks: Goal Setting, Action Planning, Prompt/Cue, and Self-monitoring.

Measures

- BCT satisfaction survey
- Co-walking report survey

Statistical Analysis

RQ1: Friedman test (non-parametric repeated measures) with post-hoc pairwise Wilcoxon signed-rank tests (Bonferroni correction); Kendall's W effect size. Mann-Whitney U test comparing BCT favorability between the 1-hour Trial and the 4-hour Trial.

RQ2: Spearman correlations for co-walking frequency and care recipient physical/emotional health; Bias-Corrected and Accelerated bootstrapped confidence intervals.

RESULTS

Table 1. Participant characteristics

Characteristic	N = 146¹
age	45.9 (13.6)
sex	
Female	118 (81%)
Male	28 (19%)
race	
Asian	4 (2.7%)
Black or African American	50 (34%)
More than one race	7 (4.8%)
Other	4 (2.7%)
White	81 (55%)
ethnicity	
Hispanic or Latino	23 (16%)
Not Hispanic or Latino	122 (84%)
Prefer not to answer	1 (0.7%)

¹Mean (SD); n (%)

Table 2. BCT Favorability

BCT	Total Sample Mean($\pm$ SD)	1-hour Trial Mean($\pm$ SD)	4-hour Trial Mean($\pm$ SD)
Goal Setting	3.50($\pm$ 0.60)	3.47($\pm$ 0.58)	3.54($\pm$ 0.62)
Action Planning	3.49($\pm$ 0.64)	3.51($\pm$ 0.64)	3.48($\pm$ 0.64)
Prompt/Cue	3.59($\pm$ 0.56)	3.57($\pm$ 0.55)	3.60($\pm$ 0.56)
Self-monitoring	3.48($\pm$ 0.66)	3.48($\pm$ 0.60)	3.49($\pm$ 0.71)

Figure 1. BCT Favorability by Trial

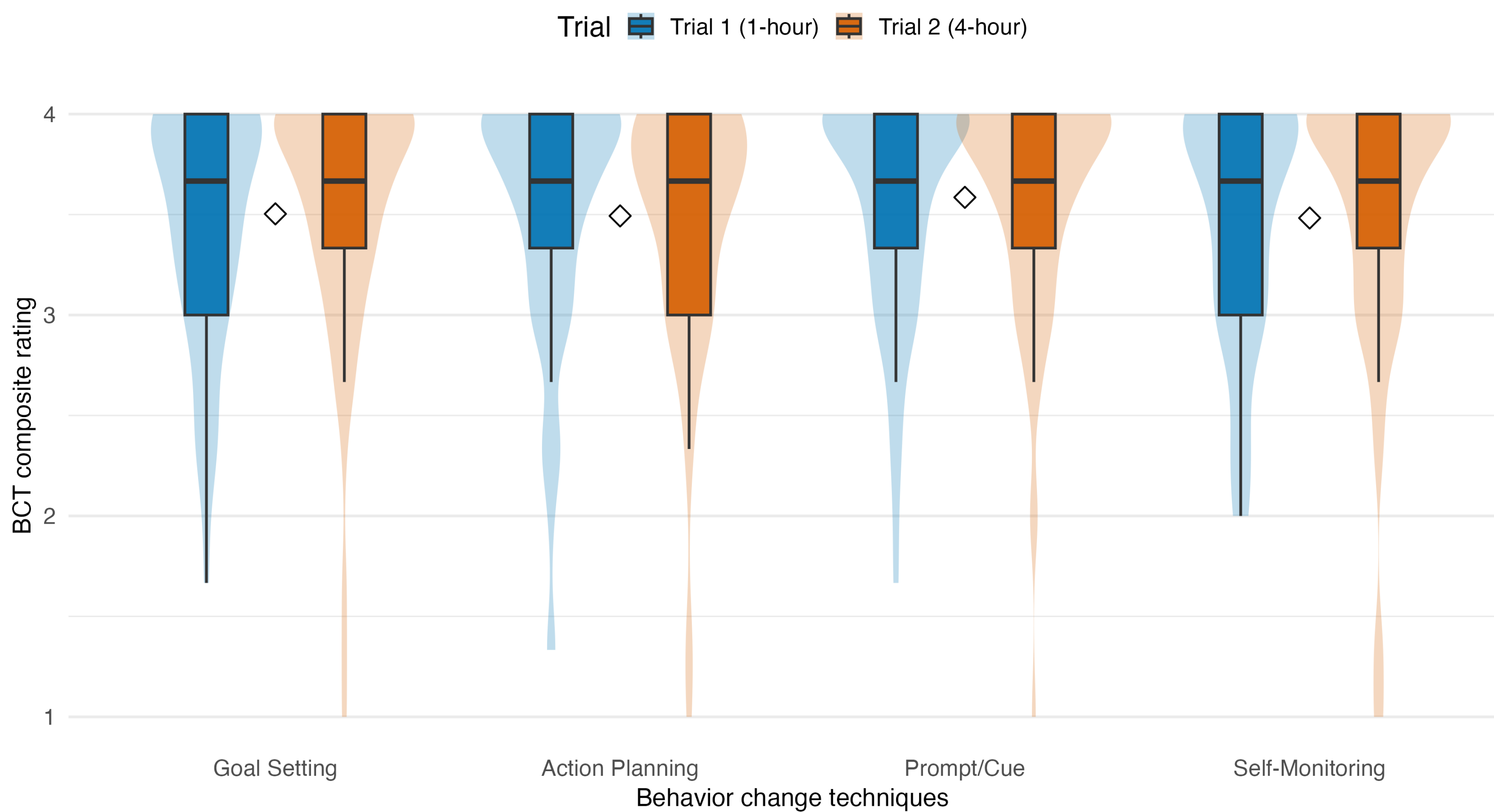
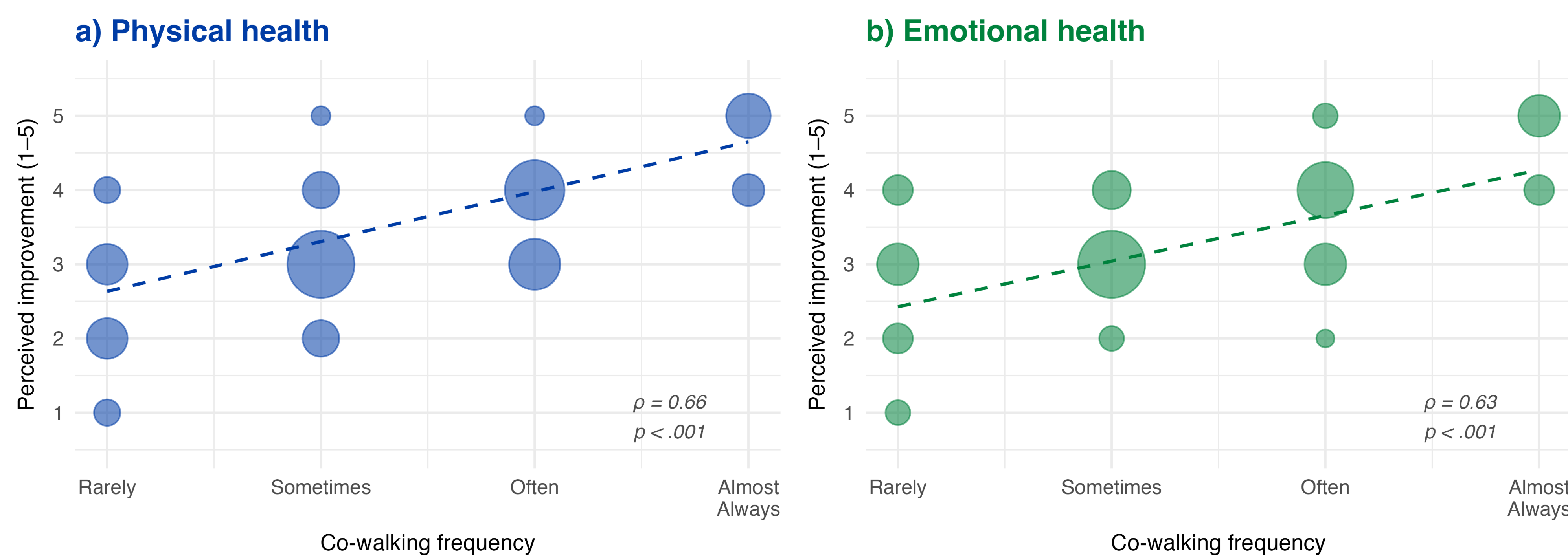


Figure 2. Association of Co-walking Frequency on Physical and Emotional Health



SUMMARY OF RESULTS

- Prompt/Cue BCT was consistently most favored by AD/ARDD caregivers across both trials (Figure 1), rated significantly higher than Action Planning ($p < 0.05$).
- No other BCTs showed significant favorability differences ($p=1.0$) (Table 2), and no difference was found between 1-hour Trial and 4-hour Trial (Figure 1).
- A strong, positive correlation existed between co-walking frequency and caregiver-perceived improvements in care recipient's physical ($p=0.66$) and emotional ($p=0.63$) health ($p < 0.001$) (Figure 2).

DISCUSSION

The Prompt/Cue BCT ratings suggest direct prompts are more favored by AD/ABDR caregivers in text-based interventions (partially supporting H1a). BCT favorability did not differ significantly across the two intervention trials (H1b not supported). The strong association of co-walking frequency with care recipient health outcomes highlights a promising intervention area (supporting H2). This suggests caregiver participation in physical activity may positively impact care recipient well-being.

CONCLUSION/FUTURE WORK

These insights will drive future research and intervention design development for caregivers and patients with AD/ADRD. The consistency of BCT favorability across both trials demonstrates that text-based interventions remain effective in both one-hour and four-hour formats, offering flexibility to accommodate caregivers' unique demands and schedules. Future research should leverage satisfaction data to optimize BCT methods and co-walking data to further investigate the therapeutic benefits of shared physical activity on patient well-being.

LIMITATIONS

The co-walking analysis was an exploratory, late-stage survey (n=86/146). Findings are caregiver perceptions, collected before full outcome analysis. Thus, results offer preliminary insights into caregivers' experiences, not definitive objective outcomes.

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